1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In breast cancer, tumor cell–derived PROS1 promotes proliferation (high MKI67) through coagulation- and microenvironment–related signaling that diminishes the prognostic value of stromal-to-tumor ratio (TSR). Consequently, in TCGA-BRCA, among tumors with low TSR, those with high PROS1 and high MKI67 will have worse overall survival than low TSR tumors with low PROS1 (and low MKI67), indicating PROS1 as a context-dependent modifier of the TSR–OS relationship.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PROS1, MKI67
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 is expressed in tumors and associates with the proliferation marker Ki-67, so why does stromal context (TSR) not consistently predict patient outcomes—could tumor-expressed PROS1-driven proliferation uncouple TSR from prognostic value? 2) Tier 1 path used: the single provided path linking tumor expression of PROS1 to Ki-67 (proliferation).

MKI67: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

3. Clinically Testable Predictions

Prediction 1: and alter stromal interactions. Mechanistic link: PROS1 signaling could modulate microenvironmental cues through coagulation-related pathways and TAM receptor–mediated signaling, potentially enhancing tumor cell proliferation in contexts where stroma is limited (low TSR).

Prediction 2: Analyze TCGA-BRCA to test for a statistical interaction between TSR and PROS1 expression, conditioning on MKI67 expression;

Prediction 3: Fit Cox models with main effects (TSR, PROS1, MKI

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions